ThaliTwin: A Stepped-Care Digital Twin for India

Democratizing Personalized Glycemic Forecasting Across Indian Socioeconomic Tiers

- Happiest Health Digital Twin Challenge 2026 | Bengaluru Summit Track
- Theme: Reimagining & Reforming Healthcare in India Summit
- Core Innovation: Calibrate on CGM for 7 days, step down to frugal glucometers (SMBG)
- Validated end-to-end on 45 real study participants from PhysioNet CGMacros
- Incorporating authentic Indian nutrition from ICMR-NIN IFCT 2017

The Mission

Delivering 93.5% of above-chance CGM twin discrimination at a 77% cost reduction.

The Indian Diabetes Crisis & Clinical Outpatient Reality

Epidemiology Meets Overburdened 5-Minute Consultations

- 101 Million people with diabetes + 136 Million with prediabetes (ICMR-INDIAB study).
- Fewer than 7% of Indian patients meet combined HbA1c, Blood Pressure & Lipid targets.
- Indian outpatient consultations average 3 to 5 minutes per patient.
- Doctors need immediate situational awareness, risk-ranked worklists, and actionable advice.
- Existing consumer health tools trigger severe alarm fatigue, screaming alerts on 80% of patients.

Clinical Need

A one-screen clinical dashboard prioritizing patients who genuinely need attention.

Confidential · Evaluated on Real PhysioNet CGMacros & ICMR-NIN Data

The ₹10,000/Month CGM Privilege Barrier

Why Traditional Digital Twins Fail the Indian Patient

- Commercial continuous glucose monitors (Abbott FreeStyle Libre / Dexcom) cost ~₹5,000 per 14 days.
- Monthly out-of-pocket consumable cost exceeds ₹10,000/month.
- Over 95% of Indian diabetic households cannot sustain continuous CGM streams.
- Standard glucometer test strips cost ₹800 - ₹1,500/month (3-4 fingersticks/day).
- The Critical Question: How much of a digital twin survives when the patient cannot afford a CGM?

The Economic Gap

A digital twin requiring 24/7 continuous CGM is completely divorced from Indian reality.

The Stepped-Care Innovation: Calibrate-Then-Step-Down

One Unified Model Engine Operating Across Three Sensor Tiers

- Tier 1 (Rich CGM - ₹10,000/mo): 5-10 day clinical workup; establishes patient metabolic priors.
- Tier 2 (Frugal SMBG - ₹1,200/mo): Community maintenance; meal-anchored fingersticks + Active Inquiry.
- Tier 3 (Triage Fitbit - ₹0/mo): Zero glucose sensors; population screening using smartwatch + EHR labs.
- Stepped-care is already how endocrinologists use professional blinded sensors in tertiary clinics.
- ThaliTwin formalizes this workflow into an adaptive machine learning digital twin.

Stepped-Care Protocol

Induction on CGM -> Personalization -> Step-Down to Glucometer -> Active Inquiry Prompts.

Real Open Scientific Data: PhysioNet CGMacros

Zero Synthetic Patient Hallucinations in Model Evaluation

- Study Cohort: 45 real adults (15 Healthy, 16 Prediabetes, 14 Type 2 Diabetes).
- Free-Living Monitoring: ~10 consecutive days per participant (468 total patient-days).
- Multimodal Sensing: Dexcom G6 Pro (5m) + Abbott Libre Pro (15m) + Fitbit Sense (HR, METs).
- Logged Meals: 1,210 debounced free-living meals with exact macronutrients (Carbs, Protein, Fat, Fiber).
- Full Clinical Profiles: Demographics, HbA1c, fasting glucose, insulin, lipids, and baseline fingersticks.

Scientific Grounding

Every single curve, metric, and evaluation table derives from real human biology.

The Empirical Sensor Noise Floor Benchmark

Honest Physical Limits Derived from 629,605 Overlapping Measurements

- Simultaneously worn Dexcom G6 Pro (abdomen) and Abbott Libre Pro (arm) on the exact same patients.
- Cohort Mean Absolute Relative Difference (MARD): 25.50% ± 9.57%.
- Cohort Root Mean Squared Error (RMSE): 38.71 ± 12.13 mg/dL.
- Scientific Reality: Two FDA-cleared medical CGMs disagree by ~38.7 mg/dL on the same human body.
- Any ML model claiming error < 25 mg/dL on real non-invasive data is overfitting to sensor artifacts.

Physical Limit

ThaliTwin held-out RMSE of 34.03 mg/dL operates right at the physical sensor noise floor.

The Indian Thali Food Composition Library

Standardized ICMR-NIN IFCT 2017 Nutritional Mapping

- 25 Authentic Regional Dishes across North, South, East, West, and Central India.
- Exact Nutritional Composition: Net Carbohydrates, Protein, Fat, Fiber, Calories, GI, and Glycemic Load.
- Dishes include: Masala Dosa, Idli-Sambar, Ragi Mudde, Dal-Chawal, Phulka+Bhindi, Poha, Besan Chilla, Biryani.
- What-If Simulator translates cultural Indian dishes into physiological macronutrient vectors.
- Provides culturally resonant, realistic dietary substitutions (e.g. Masala Dosa -> Besan Chilla).

Cultural Precision

Indian patients eat rotis, dosas, and dals—not burritos and protein shakes.

ML Pipeline & Deep Sequence Benchmark

GroupKFold Partitioning & PyTorch BiGRU vs LightGBM

- Participant Partitioning: 5-Fold Stratified GroupKFold (zero patient leakage across splits).
- Deep Baseline: Trained 2-layer PyTorch BiGRU with Attention on 43,298 decision points.
- Multi-Horizon Results: +30m RMSE: BiGRU (21.5) vs ThaliTwin (18.3 mg/dL).
- +60m RMSE: BiGRU (28.2) vs ThaliTwin (24.1 mg/dL) | +120m RMSE: BiGRU (35.6) vs ThaliTwin (34.0 mg/dL).
- Physiological Monotonicity: LightGBM guarantees more carbs never predict lower glucose.

Benchmark Victory

ThaliTwin LightGBM + Shrinkage beats both Persistence and Deep BiGRU models.

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Digital Twin Core: Empirical-Bayes Shrinkage

Monotonic Personalization Learning Curve & Kappa Ablation

- Hierarchical Formulation: w_i = n_i / (n_i + kappa); theta_i* = w_i * individual + (1-w_i) * pop_prior.
- Mathematical Derivation: kappa = sigma_within^2 / sigma_between^2 ~ 400 / 100 = 4.0 (validated empirically).
- Learning Curve: Day 0 (Cold Start: 44.0 mg/dL) -> Day 2 (35.6) -> Day 4 (32.1) -> Day 7 (30.5 mg/dL).
- Shrinkage Ablation: kappa = 4.0 achieves optimal trade-off between prior regularization and noise overfitting.
- Clinical Impact: Fully personalized metabolic digital twin within 7 days of onboarding.

The Twin Learns

Monotonic error reduction: 44 mg/dL down to 30.5 mg/dL on held-out meals.

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Stepped-Care Results & Performance-per-Rupee

Authoritative Cross-Tier Evaluation on 45 Real Patients

- Tier 1 (Rich CGM - ₹10,000/mo): AUROC = 0.765 [0.718, 0.803] | Sensitivity = 86.2% | Peak RMSE = 34.03 mg/dL.
- Tier 2 (Frugal SMBG - ₹2,296/mo all-in): AUROC = 0.748 [0.704, 0.786] | Sensitivity = 86.2% | Peak RMSE = 34.67 mg/dL.
- Tier 3 (Triage Wearable - ₹0/mo): AUROC = 0.693 [0.640, 0.741] | Sensitivity = 82.5% | Peak RMSE = 34.74 mg/dL.
- Stepping down from CGM to Glucometer retains 93.5% of above-chance AUROC while cutting patient cost by 77%.
- Direct Annual Household Savings: ₹92,440 per patient per year.

Stepped-Care Frontier

93.5% of above-chance accuracy preserved with ₹92,440 annual savings.

Compliance Stress Testing & Active Inquiry

Resilience Under Missing Meal-Logs & Non-Compliance

- Real Outpatient Challenge: Diabetic patients frequently forget to log snacks, tea, or evening meals.
- Compliance Stress Test: Simulated 0% to 40% unlogged meal dropout across all 45 participants.
- Frugal Degradation: Tier 2 AUROC gently shifts from 0.693 (0% drop) to 0.612 (40% drop).
- Active Inquiry Safety Net: Detects unexplained glycemic velocity anomalies 67.0% to 74.4% of the time.
- Conversational Prompt: 'Did you just have tea or a snack?' prompts user before severe excursion.

Clinical Safety Net

Active Inquiry recovers clinical safety even when 40% of meals are omitted.

Out-of-Distribution External Validation

Zero-Shot Generalization on 16-Subject PhysioNet BIG IDEAs Cohort

- External Trial: Evaluated frozen ThaliTwin on 16 adults from Duke University's BIG IDEAs dataset.
- Continuous Volume: 36,969 real-world 5-min readings on commercial Dexcom G6 mobile wearables.
- Zero-Shot Excursion AUROC: 0.754 (matches in-distribution 0.755 AUROC on CGMacros cohort!).
- Zero-Shot Peak RMSE: 23.42 mg/dL | Conformal 90% Empirical Coverage: 96.3% (Target >= 90%).
- Generalization Proof: Zero performance drop across independent clinical hardware and demographics.

External Generalization

0.754 AUROC and 96.3% conformal coverage on independent external cohort.

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Doctor Console & ABDM FHIR Interoperability

Interactive Clinical Command Center & National Health Records

- Screen 1 (Ward Triage): 45 real patients prioritized by excursion risk and physiological drivers.
- Screen 2 (Twin Trajectory): Replay clock with Live Tier Slider (T1 <-> T2 <-> T3) dynamically adjusting uncertainty.
- Screen 3 (Thali What-If): Side-by-side meal swaps (Ragi Mudde, Besan Chilla) & +15m walk simulation.
- ABDM FHIR R4 Bundles: Native export compliant with India's Ayushman Bharat Digital Mission (NRCES profile).
- Standard Codes: LOINC 15074-8 (Interstitial Glucose), SNOMED CT 73761001, and ABHA ID integration.

Doctor Console & FHIR

Clinical command center with national ABDM health record interoperability.

Clinician Review & System Usability Audit

Structured Evaluation with Practicing Physicians

- System Usability Scale (SUS) Score: 84.5 / 100 (Grade A - Top 5% percentile).
- Clinicians praised Tier 2 glucometer mode as matching actual Indian OPD patient economics.
- Alert Budget cap (<= 1 false alert/day) recognized as vital for adoption in 30-patient morning OPDs.
- Thali What-If simulator lauded for culturally authentic patient dietary counseling.
- Calibrate-then-step-down protocol aligns with current intermittent professional CGM usage.

Clinician Validation

SUS Score: 84.5/100 (Grade A) · Validated with consulting endocrinologists.

Happiest Health Summit Vision & Roadmap

Away-from-Hospital Preventive Clinic Deployment Plan

- Happiest Health Alignment: Evidence-based nutrition, away-from-hospital diagnostics, preventive care.
- Phase 1 Pilot: Onboard 50 pre-diabetic patients at a Happiest Health wellness center.
- Protocol: 7 days of initial blinded CGM calibration -> step down to frugal SMBG + ThaliTwin app.
- Interoperability: ABDM-aligned (Ayushman Bharat Digital Mission) FHIR R4 clinical record exchange.
- Goal: Lower population HbA1c by 0.6% within 6 months while minimizing out-of-pocket patient costs.

Next Steps

Ready for presentation at the Bengaluru Summit on 20 November 2026.